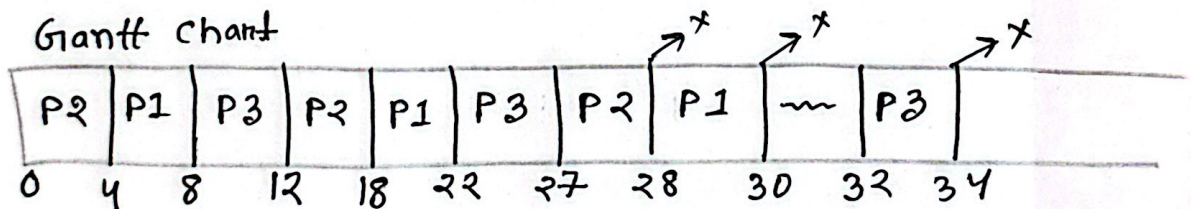


Fall 2025 (A) → CPU Scheduling

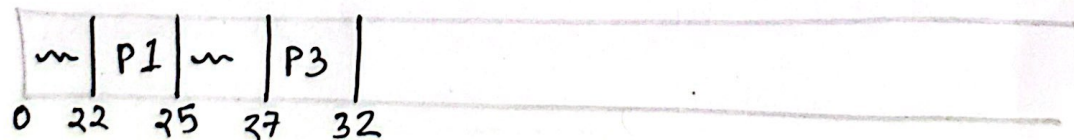
Q) b) (i)

Process	Arrival time	Burst time	I/O time
P1	4	10 8 7 0 8 4 0	3s I/O (after total 8s CPU)
P2	0	4 7 2 0	N/A
P3	6	4 7 2 0 9 5 0	5s I/O (after total 9s CPU)

Q0 (RR: 4)	Q1 (RR: 6)	Q2 (RR: 8)	I/O
0: P2	4: P2	18: P2	23: P1
4: P1	8: P2, P1	22: P2	25: { }
6: P3	12: P2, P1, P3	27: P2	27: P3
12: { }	18: P1, P3	28: { }	32: { }
25: P1	22: P3		
32: P1, P3	27: { }		
34: { }			



I/O Granff Chart



(ii)

Process	Turnaround time	Waiting time
P1	$30 - 4 = 26$	$26 - 10 - 3 = 13$
P2	$28 - 0 = 28$	$28 - 11 = 17$
P3	$34 - 6 = 28$	$28 - 11 - 5 = 12$
	average = 27.33	average = 14

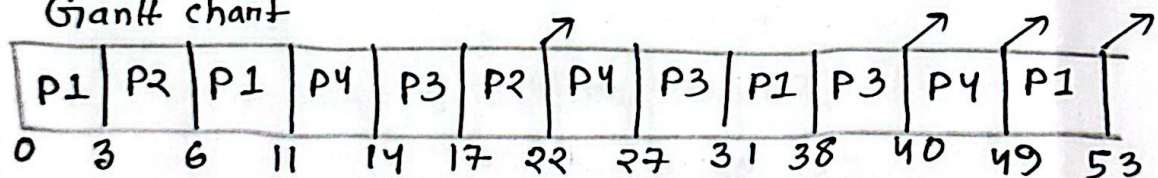
Spring 25(A)

① c) (i)

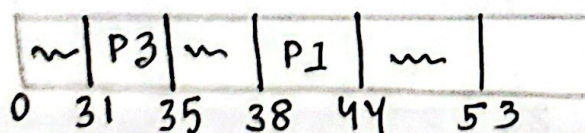
Process	Arrival time	Burst time	I/O time
P1	0	15 16 11 40 15 17 70	6 (after 153 CPU)
P2	2	8 5 0	N/A
P3	12	8 6 20 7 40	4 (after 75 CPU)
P4	7	17 14 90	N/A

Q1 (RR: 3)	Q2 (RR: 5)	Q3 (RR: 10)	I/O
0: P1	3: P1	11: P1	31: P3
2: P2	6: P1, P2	14: P1	35: { }
7: P4	11: P2	22: P1, P2	38: P1
12: P3	14: P2, P4	27: P1, P4	44: { }
17: { }	17: P2, P4, P3	31: P1, P4	
35: P3	22: P4, P3	38: P4	
44: P3 P1	27: P3	49: { }	
53: { }	31: { }		

Gantt chart



I/O Gantt chart



(ii)

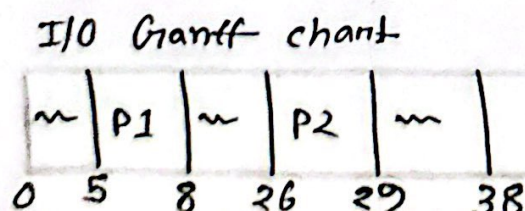
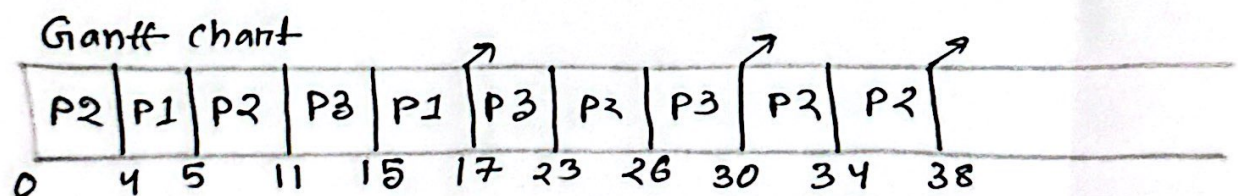
Process	Turnaround time	Waiting time
P1	$53 - 0 = 53$	$53 - 19 - 6 = 28$
P2	$22 - 2 = 20$	$20 - 8 = 12$
P3	$40 - 12 = 28$	$28 - 9 - 4 = 15$
P4	$49 - 7 = 42$	$42 - 17 = 25$
	average = 35.75	average = 20

Summer 2025 (A)

(2) b) (i)

Process	Arrival time	Burst time	I/O time
P1	3	2 2 0 2 0	3 (after 13s of CPU)
P2	0	2 1 1 1 1 4 1 3 1 1 0	3 (after 13s of CPU)
P3	7	1 1 1 0	N/A

Q ₀ (RR: 4)	Q ₁ (RR: 6)	Q ₂ (RR: 8)	I/O
0: P2	4: P2	11: P2	5: P1
2: P1	5: P2	15: P2	8: { }
5: { }	11: { }	17: P2	26: P2
7: P3	15: P3	23: P2, P3	29: { }
8: P3, P1	17: P3	26: P3	
15: P1	23: { }	30: { }	
17: { }	34: P2		
29: P2	38: { }		
34: { }			



(ii)

Process	Turnaround time	Waiting time
P1	$17 - 2 = 15$	$15 - 3 - 3 = 9$
P2	$38 - 0 = 38$	$38 - 21 - 3 = 14$
P3	$30 - 7 = 23$	$23 - 14 = 9$
	Average = 25.33	Average = 10.67